SYSC 5108 Exam Study
Assignment 2, Question 1

Jesse Levine

Question

Assume that the prior for q, the proportion of spam emails sent, is a random variable with a uniform distribution. Let x denote the number of spam emails received in a random sample of n emails. Find the posterior distribution $p(q \mid x)$. Which distribution will $p(q \mid x)$ be?

Course and Textbook Cross-References

- **Chapter 3 slides, Probability Review:** use Bayes' formula, $p(X \mid Y) = p(X)p(Y \mid X)/p(Y)$, with the denominator acting as a normalizing constant.
- **Chapter 3 slides, Maximum Likelihood Estimator:** the likelihood is the probability of the observed sample as a function of the unknown parameter. Here that parameter is the spam probability q.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3:** a Bernoulli random variable is binary and has $P(X = 1) = \phi$, $P(X = 0) = 1 - \phi$. A spam/not-spam email is therefore a Bernoulli trial with parameter q, and x spam emails in n trials gives a binomial likelihood.
- **Goodfellow et al., Chapter 5, Bayesian Statistics:** before observing data, knowledge about an unknown parameter is represented by a prior $p(\theta)$; after observing data, the likelihood is combined with the prior using Bayes' rule to obtain the posterior $p(\theta \mid \text{data})$.

Step-by-Step Solution

1. Identify the random variables

Each email has two possible outcomes:

$$\text{spam} = 1, \qquad \text{not spam} = 0.$$

If q is the unknown probability that an email is spam, then each email can be modeled as a Bernoulli trial:

$$Y_i \mid q \sim \text{Bernoulli}(q).$$

The observed statistic is the count of spam emails:

$$x = \sum_{i=1}^{n} Y_i.$$

Therefore, conditional on q, the count x follows a binomial distribution:

$$X \mid q \sim \text{Binomial}(n, q).$$

2. Write the likelihood

The binomial likelihood is

$$p(x \mid q) = \binom{n}{x} q^x (1-q)^{n-x}, \qquad 0 \le q \le 1.$$

When treating this as a function of q, the binomial coefficient $\binom{n}{x}$ is constant because n and x are already observed.

3. Write the prior

The question states that q has a uniform prior on $[0, 1]$:

$$q \sim \text{Uniform}(0, 1).$$

This has density

$$p(q) = 1, \qquad 0 \le q \le 1.$$

Equivalently,

$$\text{Uniform}(0, 1) = \text{Beta}(1, 1).$$

4. Apply Bayes' rule

Bayes' rule gives

$$p(q \mid x) = \frac{p(x \mid q)p(q)}{p(x)}.$$

The denominator $p(x)$ does not depend on q, so it only normalizes the posterior. Using proportionality,

$$p(q \mid x) \propto p(x \mid q)p(q).$$

Substitute the likelihood and prior:

$$p(q \mid x) \propto \binom{n}{x} q^x (1-q)^{n-x} \cdot 1.$$

Remove the constant $\binom{n}{x}$, since it does not depend on q:

$$p(q \mid x) \propto q^x (1-q)^{n-x}.$$

5. Match the kernel to a Beta distribution

A Beta distribution with parameters α and β has density

$$\text{Beta}(q; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} q^{\alpha-1} (1-q)^{\beta-1}, \qquad 0 \le q \le 1,$$

where

$$B(\alpha, \beta) = \int_0^1 q^{\alpha-1} (1-q)^{\beta-1} \, dq$$

is the Beta function.

Compare the posterior kernel

$$q^x(1-q)^{n-x}$$

with the Beta kernel

$$q^{\alpha-1}(1-q)^{\beta-1}.$$

Matching exponents gives

$$\alpha - 1 = x, \qquad \beta - 1 = n - x.$$

Therefore,

$$\alpha = x + 1, \qquad \beta = n - x + 1.$$

Final Answer

$$\boxed{p(q \mid x) = \mathrm{Beta}(q; x+1, n-x+1)}$$

or, written as a normalized density,

$$\boxed{p(q \mid x) = \frac{1}{B(x+1, n-x+1)} q^x(1-q)^{n-x}, \qquad 0 \le q \le 1.}$$

The posterior is a **Beta distribution**. In words, the uniform prior $\mathrm{Beta}(1,1)$ contributes one pseudo-count to each class, so after observing x spam emails and $n-x$ non-spam emails, the posterior becomes

$$q \mid x \sim \mathrm{Beta}(1+x,\ 1+n-x).$$

Exam Notes

- This is the standard Beta-binomial conjugacy result:

$$\mathrm{Beta}(\alpha, \beta) + \mathrm{Binomial}(n, q) \to \mathrm{Beta}(\alpha + x, \beta + n - x).$$

- With the assignment's uniform prior, $\alpha = \beta = 1$, so the result reduces to $\mathrm{Beta}(x+1, n-x+1)$.
- The posterior mean is

$$\mathbb{E}[q \mid x] = \frac{x+1}{n+2}.$$

  This is not required to answer the question, but it is a useful exam check: it is a smoothed estimate of the spam proportion, slightly pulled away from exactly 0 or 1 by the uniform prior.